EC 1745 Practice Midterm (mock exam for Fall 2026), v3

This is not an official exam. It is written in the style of the Fall 2024 and Fall 2025 midterms and covers chapters 1–6 only (no stock valuation). It tests the topics your professor listed: building financial statements, deriving cash flows, ratios, valuing and restructuring commitments, bond issuance and restructuring, and duration.

Rules to practice under: start from a **blank** Excel workbook. Put each part on its own tab. Use only your one-page notes sheet. Suggested time: **70 minutes**. The real exam is 60 minutes and will probably be shorter than this one.

All dollar amounts are in $ millions unless noted. The answer key is in **EC1745_Mock_Midterm_ANSWER_KEY_v3.xlsx** (live formulas), and a one-page answer list is on the last page of this document.

Part 1. Financial statements and ratios: Pemberton Supply Co.

Pemberton Supply Co. reported the following for fiscal 2025 (year ended December 31):

Revenue 2,400 · Cost of goods sold 1,440 · SG&A 480 · Depreciation 120 · Interest expense 40 · Tax rate 25% · Dividends paid 60

Balance sheet item	12/31/2024	12/31/2025
Cash	50	?
Accounts receivable	200	230
Inventory	250	280
Net PP&E	1,000	1,080
Accounts payable	150	170
Short-term debt	100	100
Long-term debt	400	450
Shareholders' equity	850	?

Pemberton issued no shares and repurchased none in 2025. It has 50 million shares outstanding at $40 per share. Treat all cash as excess cash.

A. Construct the 2025 income statement. Report gross profit, EBITDA, EBIT, and net income.

B. What is shareholders' equity at 12/31/2025?

C. What 12/31/2025 cash balance makes the balance sheet balance? What are total assets?

D. Calculate 2025 cash flow from operations, capital expenditures, and free cash flow. Show that CFO + CFI + CFF equals the change in cash. Then use the textbook's method: operating cash flow, net capital spending, change in net working capital, and cash flow from assets. Show that cash flow from assets equals cash flow to creditors plus cash flow to stockholders.

E. Calculate gross margin and ROE (use year-end equity). Decompose ROE with the DuPont identity.

F. Calculate the sustainable and internal growth rates.

G. Calculate the current ratio, interest coverage (EBIT / interest), and book debt-to-equity ratio.

H. Calculate market capitalization, enterprise value, EV/EBITDA, and the P/E ratio.

Part 2. Valuing and restructuring a commitment: Corrigan Steel

Corrigan Steel has agreed to settle a lawsuit by paying the plaintiff $10 million per year for 15 years, with the first payment one year from today. Use a discount rate of 8% per year unless told otherwise.

A. What is the present value of the settlement? Do it once with a timeline (the “dumb” way) and once with an Excel function.

B. Corrigan proposes instead to pay $7 million in year 1, growing 4% per year, for 15 years. What is its present value? Which plan should the plaintiff prefer?

C. The plaintiff asks instead for a single payment of $90 million today. What discount rate makes $90 million today worth the same as the original 15 payments? At 8%, which should Corrigan prefer?

D. Suppose the 15 payments started today instead (an annuity due). What level annual payment has the same present value as the original plan?

E. To pay $90 million today, Corrigan would take a 10-year bank loan at 7.2% APR with monthly payments. What is the monthly payment? What is the loan's effective annual rate?

F. What is the loan balance immediately after the 36th monthly payment?

Part 3. Bond restructuring with no dates: Brewster Corp.

Two years ago Brewster Corp. issued 10-year bonds at par with a 5% annual coupon paid semiannually. $400 million of face value is outstanding. Brewster's bonds now yield 7% (APR, semiannual compounding).

A. What is the price of a Brewster bond today, per $100 of face value?

B. What is the bond's effective annual yield?

C. What is the market value of all of Brewster's outstanding bonds?

D. Brewster proposes to exchange the old bonds for new bonds with the same face value that mature 10 years from today and are priced to yield 7%. What annual coupon rate must the new bonds pay for bondholders to be indifferent?

E. Build the old bond's cash-flow timeline. Calculate its Macaulay duration (in years) and modified duration. Check your answer with Excel's DURATION function.

F. Use modified duration to estimate the percent change in the old bond's price if its yield rises from 7% to 7.5%. Then calculate the actual new price and percent change. Why do they differ?

Part 4. Bond pricing with dates: Hollis Outdoor Inc.

Use a settlement date of **10/13/2026** for every part.

Hollis Outdoor has a bond that matures on 11/15/2036. It pays a 6.25% annual coupon semiannually, accrues interest on a 30/360 basis (Excel basis 0), and pays 100% of par at maturity.

The benchmark Treasury matures on 8/15/2036. It has a 4.5% coupon paid semiannually, a clean price of 101.25, and ACT/ACT accrual (Excel basis 1). Hollis's spread over this Treasury is 150 basis points.

A. What is the Treasury's yield to maturity?

B. What is the Hollis bond's yield?

C. What is the Hollis bond's clean price (per $100 par)?

D. How many days of the current coupon period have passed? How many days are in the period? What is the accrued interest?

E. What price would a dealer quote you? What price would you actually pay?

F. What are the bond's Macaulay duration and modified duration?

G. Use modified duration to estimate the percent change in the dirty price if the yield falls by 50 basis points. Then calculate the actual change.

H. Hollis wants to raise $250 million with new 10-year bonds at the same yield. (a) What coupon rate lets the bonds sell at par? (b) If Hollis instead sets a 5% coupon, what total face value must it issue? Treat the new bonds as issued today with exactly 20 semiannual periods.

Answers (check after you finish)

Question	Answer
1.A Gross profit, EBITDA, EBIT, net income	960; 480; 360; 240
1.B Equity 12/31/2025	1,030
1.C Cash (plug); total assets	160; 1,750
1.D CFO; CapEx; FCF	320; 200; 150. Check: 320 + (−200) + (−10) = 110 = ΔCash
1.D Textbook: OCF; net capital spending; ΔNWC; CF from assets	400; 200; 150; 50 = creditors (−10) + stockholders (60)
1.E Gross margin; ROE	40.00%; 23.30%
1.E DuPont: margin × turnover × multiplier	10.00% × 1.3714 × 1.6990
1.F Sustainable; internal growth (b = 0.75)	21.18%; 11.46% (textbook formula). Simpler ROE × b form: 17.48%; 10.29%
1.G Current ratio; coverage; book D/E	2.48; 9.0x; 0.534 (0.699 counting all liabilities as debt, as the textbook does)
1.H Market cap; EV; EV/EBITDA; P/E	2,000; 2,390; 4.98x; 8.33x
2.A PV of settlement	85.595
2.B PV of growing plan	75.647. Plaintiff prefers the original plan
2.C Implied rate for $90m lump sum	7.19%. At 8%, Corrigan keeps paying annually (85.6 < 90)
2.D Annuity-due payment	9.259 (= 10 / 1.08)
2.E Monthly payment; EAR	1.0543; 7.442%
2.F Balance after 36 payments	69.403
3.A Price per 100	87.906
3.B Effective annual yield	7.1225%
3.C Market value of debt	351.62
3.D Coupon rate on new bond	5.298%
3.E Macaulay; modified duration	6.585 years; 6.362
3.F Estimated; actual % change (new price)	−3.181%; −3.121% (85.162). Gap is convexity
4.A Treasury YTM	4.342%
4.B Hollis YTM	5.842%
4.C Clean price	103.072
4.D Days elapsed; days in period; AI	148; 180; 2.569
4.E Dealer quote; price paid	103.072 (clean); 105.641 (dirty)
4.F Macaulay; modified duration	7.490; 7.277
4.G Estimated; actual % change	+3.639%; +3.725%
4.H (a) Par coupon; (b) face for 5% coupon	5.842%; 266.83 (price 93.691)